

# EchoLoc: Audio-Aware Object Grounding via Joint Heatmap and Box-Level Localization

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## Abstract

Although recent multimodal models aim to jointly understand images, text, and audio, they mostly focus on global-level representations and remain limited in precise object-level grounding. In particular, conventional text-image grounding models struggle to leverage non-visual cues such as sound. We propose an audio-aware grounding framework called *EchoLoc*, based on a query-based transformer detector of MDETR, incorporating audio information to complement textual supervision and localize sound-related visual regions. Our approach learns spatial heatmap-based local cues through audio-visual alignment and uses them to guide ROI-level object box prediction, enabling the localization of sound-related visual regions even without explicit bounding-box annotations. On Flickr-SoundNet-Test, *EchoLoc* yields +2.58% and +17.08% relative improvements in cIoU (adap) and AUC (adap) over the state-of-the-art.

**Index Terms:** Audio-visual grounding, Sound source localization, Multimodal representation learning

## 1. Introduction

Recent technology in artificial intelligence has expedited multimodal learning. By integrating semantically aligned features from multimodal datasets, models can understand images, text, and audio within a unified framework. However, most models [1, 2] mainly focus on global correspondence and semantics, but lack fine-grained object-level detection and localization.

To tackle the object-level localization, vision-language grounding has made some considerable advances. By aligning natural language with visual regions, some representative works [3, 4] localize objects by predicting bounding boxes under an open-vocabulary setting. However, these methods rely heavily on complete and semantically explicit captions, making them sensitive to noisy or incomplete textual signals such as those produced by automatic speech recognition (ASR).

Sound Source Localization (SSL) aims to localize the image region corresponding to the sound in paired image-audio samples. Since datasets with ground-truth sound source regions are scarce, most previous studies [5, 6, 7, 8] rely on self-supervised learning. Typically, contrastive learning approaches learn localization cues in the form of audio-visual similarity heatmap, which provides soft spatial evidence of sound source. However, such heatmaps tend to be blurry, making it difficult to obtain precise object-level bounding boxes. Moreover, because converting a heatmap into a bounding box typically depends on

post-processing, which is not included in the training objective, it is difficult to optimize end-to-end.

These limitations highlight a missing training signal that connects sound cues to object-level predictions. Although audio-visual alignment provides weak spatial cues, directly learning object boxes with clear boundaries remains difficult without box annotations. This raises a fundamental question: *How can we incorporate a pipeline for generating object candidates and selecting the sound-producing objects among them into end-to-end training, using only limited audio-visual signals?* In other words, we need a framework that predicts object boxes without additional box annotations, while preserving local, object-level grounding beyond global semantic alignment.

To address the problem of audio-aware object grounding, we present *EchoLoc*, which jointly trains heatmap prediction and box prediction modules. The heatmap module is trained via spatial and global contrastive learning. To effectively train the box prediction module, datasets with bounding box annotations corresponding to sounds are required, but they are very scarce. To tackle scarcity, we generate a pseudo bounding box from the audio-visual similarity map obtained from the heatmap module and use it as supervision for training the box prediction module.

Despite this effort, training with pseudo boxes can lead to a bootstrap paradox, because it becomes unstable for a model to learn from its own predictions. We adopt a query-based detection model to generate diverse audio-conditioned candidate boxes and select the one with the highest audio-visual similarity as the final box prediction. In this way, we do not treat the pseudo box as a hard target; instead, it encourages exploration during candidate generation and yields more stable predictions by leveraging heatmap-based signals. As such, the heatmap and box prediction modules can be co-trained in a complementary manner, allowing the box predictor to refine heatmap-derived localization cues into object-level predictions.

In summary, our contributions are threefold:

- We propose an audio-aware grounding framework, *EchoLoc*, based on a query-based transformer detector of MDETR [3], incorporating audio information to complement textual supervision and localize sound-related visual regions.
- Our approach learns spatial heatmap-based local cues through audio-visual alignment and then uses them to guide ROI-level object box prediction, enabling fine-grained localization of sound-related visual regions even without explicit bounding-box annotations.
- *EchoLoc* achieves 36.36 cIoU and 38.28 AUC on VGG-SS, and shows strong adaptive localization performance on Flickr-SoundNet-Test, yielding +2.58% and +17.08% relative improvements in cIoU(adap) and AUC(adap) over the state-of-the-art.

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## 2. Methods

### 2.1. Architecture

We propose *EchoLoc*, an audio-aware object grounding framework that jointly performs heatmap-based and box-level localization, as illustrated in Figure 1. Each input sample consists of a single image frame and an audio signal generated within the same scene. Two modalities are independently encoded into feature representations using modality-specific encoders. For the visual stream, the input image is transformed into an object-centric visual feature map using a ResNet-101 backbone initialized with weights pre-trained in MDETR [3].

For the audio stream, we adopt Pretrained Audio Neural Networks (PANNs) [9] and use a frozen CNN14 as the audio backbone to encode the input audio into a global feature representation. We avoid heavier transformer-based audio encoders such as AST [10] due to their parameter overhead and computational complexity. The resulting visual feature map and audio feature are then fed into subsequent modules. The overall framework consists of two parallel modules that share the same input features.

The box prediction module is an audio-conditioned, object-centric branch based on the transformer architecture employed in MDETR [3]. The visual feature map and the global audio feature are jointly fed into the transformer encoder, enabling an early fusion mechanism where cross-modal self-attention inherently produces audio-aware visual features. This allows the model to integrally capture visual context related to the sound. The transformer decoder then processes object queries by attending to the fused audio-visual features and directly outputs the bounding box coordinates corresponding to the sound-producing object within the image. This module is responsible for accurately estimating object boundaries with clear visual structures.

The heatmap prediction module is a localization component that estimates sound-generating regions. We compute an audio-visual similarity heatmap by measuring pixel-wise similarity between the audio features and individual spatial locations in the visual feature map. The heatmap provides soft spatial evidence of sound source locations and is later used to generate pseudo boxes for training the box prediction module.

### 2.2. Training

To train the model to localize the sound-producing regions effectively and encourage more object-centric bounding boxes, we leverage both heatmap prediction and box prediction modules. The heatmap module is trained through contrastive learning on image-audio pairs. In particular, we utilize both spatial contrastive learning and global contrastive learning. Bounding-box annotations aligned with the audio are required to train the box prediction module, but such datasets are scarce. Therefore, we adopt a self-supervised strategy to generate pseudo boxes from the audio-visual similarity map predicted by the heatmap module and to use them to supervise the box prediction module.

#### 2.2.1. Spatial contrastive loss

We use  $s_{\text{spa}}$  as a spatial similarity for localization, which encourages the model to highlight sound-producing regions.  $s_{\text{spa}}(i, j)$  is defined as below:

$$s_{\text{spa}}(i, j) = \log \left( \sum_{x=1}^H \sum_{y=1}^W \exp(\langle \mathbf{V}_i(x, y), \mathbf{a}_j \rangle) \right) \quad (1)$$

where  $\mathbf{V}_i$  denotes the visual feature map of the  $i$ -th image sample,  $\mathbf{V}_i(x, y)$  denotes the visual feature vector at spatial location  $(x, y)$ , and  $\mathbf{a}_j$  is the audio feature vector of the  $j$ -th audio sample.  $\langle \cdot, \cdot \rangle$  denotes the inner product of two feature vectors. Since  $s_{\text{spa}}(i, j)$  aggregates spatial similarities via log-sum-exp pooling, increasing the similarity at a few highly relevant locations contributes more than uniformly increasing similarities across the entire image. It encourages the model to focus on regions that correspond to the sound source.

The contrastive loss based on  $s_{\text{spa}}$  is defined as below:

$$\mathcal{L}_{\text{spa}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s_{\text{spa}}(i, i)/\tau)}{\sum_{j=1}^B \exp(s_{\text{spa}}(i, j)/\tau)} \quad (2)$$

where  $B$  is a batch size. This loss increases the similarity of positive pairs while decreasing that of negative pairs.

The above formulation is a one-way contrastive loss using the image as an anchor. In practice, we additionally compute the loss in the reverse direction and use the average of the two losses as the final loss.

#### 2.2.2. Global contrastive loss

We use  $s_{\text{glob}}$  as a global similarity for semantic feature alignment, which aligns the overall scene semantics between audio and vision, as in previous study [8].  $s_{\text{glob}}(i, j)$  is defined as below:

$$\bar{\mathbf{v}}_i = \frac{1}{HW} \sum_{x=1}^H \sum_{y=1}^W \mathbf{V}_i(x, y); \quad s_{\text{glob}}(i, j) = \langle \bar{\mathbf{v}}_i, \mathbf{a}_j \rangle \quad (3)$$

where  $\bar{\mathbf{v}}_i$  denotes the globally pooled visual feature vector of the  $i$ -th image sample, which is computed by spatial averaging of the visual feature map  $\mathbf{V}_i$ .  $s_{\text{glob}}(i, j)$  is obtained by simple inner product of the two feature vectors. Unlike  $s_{\text{spa}}$ , which emphasizes spatial localization,  $s_{\text{glob}}$  aligns audio and vision at the scene level by matching globally pooled representations.

We then apply the same contrastive learning objective as in  $\mathcal{L}_{\text{spa}}$ , replacing  $s_{\text{spa}}$  with  $s_{\text{glob}}$  to obtain  $\mathcal{L}_{\text{glob}}$  (and similarly average the two directions).

#### 2.2.3. Box prediction loss

Now we train the detection head for bounding box prediction, using pseudo boxes generated from the heatmap prediction module. Specifically, for each image, we first identify a region that covers the top- $\alpha$  probability mass of the audio-visual similarity map, and then convert the tight bounding box of this region into a pseudo ground-truth box  $\mathbf{b}_i = (c_x, c_y, w, h)$  in normalized coordinates. To obtain the optimal matching between the model-predicted query boxes  $\{\mathbf{b}_q\}_{q=1}^N$  and the pseudo boxes, we apply the Hungarian matching, as in DETR [11]. The matching cost follows MDETR [3] and is based on a weighted sum of an audio-conditioned classification cost and box costs. Once the matching is completed, we compute the classification and box regression losses for the matched pairs. For the classification loss, we follow MDETR and apply a cross-entropy loss on the per-query classification logits; however, we replace MDETR’s original formulation for text-token alignment with an audio-conditioned one. The box regression loss is also defined as in MDETR, using a linear combination of the L1 loss and the Generalized IoU (GIoU) loss.

Finally, we use the total loss  $\mathcal{L} = \mathcal{L}_{\text{spa}} + \mathcal{L}_{\text{glob}} + \mathcal{L}_{\text{box}}$ , combining spatial localization, global semantic alignment, and box localization.

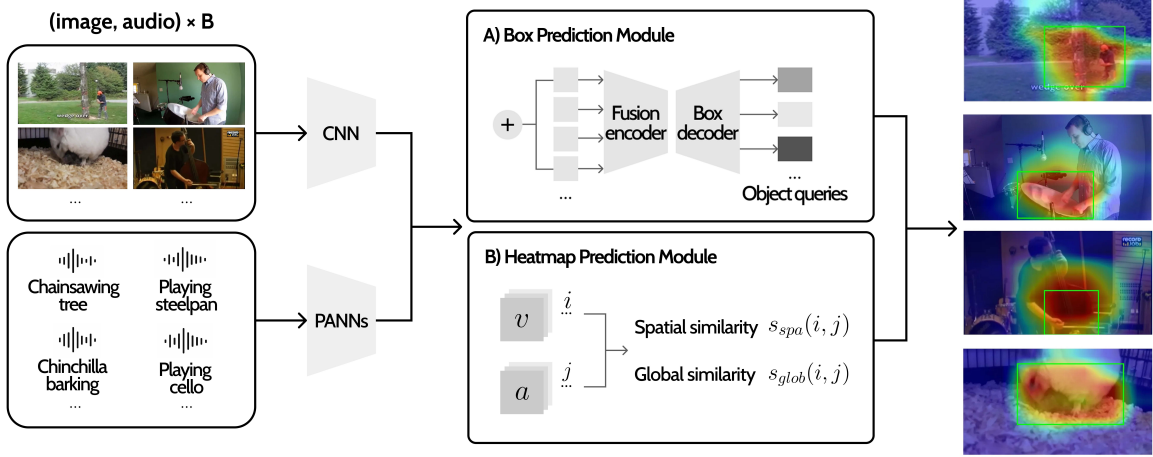


Figure 1: Overall architecture of *EchoLoc*: (1) An image–audio pair is encoded by modality-specific encoders; (2) The Box Prediction Module fuses both modalities via a Fusion encoder and uses object queries in a Box decoder to predict audio-guided bounding boxes; (3) The Heatmap Prediction Module generates sound-source heatmaps via audio–visual similarity; and (4) Two modules jointly enable complementary hard and soft grounding

### 3. Experiments

#### 3.1. Experimental setup

For training, we use two audio-visual datasets, VGGSound [12] and Flickr-SoundNet-144K [13], which are widely adopted in prior AVSL studies. Specifically, we leverage VGGSound, containing approximately 200k video clips, and construct 144k image–audio pairs from Flickr-SoundNet-144K. For evaluation, we adopt VGG-SS [6] and Flickr-SoundNet-Test [5], both of which provide manually annotated sound source labels. Each evaluation set contains approximately 5,000 image–audio pairs with bounding-box annotations indicating sound-generating regions in the visual frame.

We train our model using a batch size of 32 on a single NVIDIA GeForce RTX 3090 GPU (24 GB VRAM). All experiments are optimized using the AdamW optimizer [14]. The classification, bounding box regression, and GIoU losses are weighted by 0.5, 2.5, and 1.0, respectively. We set the pseudo mass threshold to 0.4 and the heatmap temperature parameter to 0.15. The model is trained for 40 epochs, which takes approximately 15 hours.

#### 3.2. Audio-based visual grounding

In this section, we quantitatively compare *EchoLoc* with prior sound source localization methods on VGG-SS, Flickr-SoundNet-Test benchmarks as presented in Table 1. We train on VGGSound and evaluate on VGG-SS, and train on Flickr-SoundNet-144K and evaluate on Flickr-SoundNet-Test. We report results of prior methods trained with the same amount of data for fair comparison.

Following prior related work, we report cIoU and AUC. Since fixed-threshold cIoU can be sensitive to selection of threshold, we additionally report adaptive metrics proposed in [15], including cIoU (adap) and AUC (adap).

As shown in Table 1, *EchoLoc* achieves 36.36 cIoU and 38.28 AUC on VGG-SS. Although *EchoLoc* is not state-of-the-art on all fixed-threshold metrics, these results indicate that the proposed framework remains effective for audio-aware object grounding without explicit bounding-box annotations. On

Flickr-SoundNet-Test, *EchoLoc* achieves 84.34 cIoU and shows strong adaptive performance, with 88.22 cIoU (adap) and 79.99 AUC (adap). The lower fixed-threshold AUC may be partly explained by calibration sensitivity and differences in the heatmap scale or distribution, while the adaptive results still show strong gains in ambiguous multi-instance scenes.

The results show that *EchoLoc* achieves its strongest relative gains on the Flickr-SoundNet dataset, particularly under adaptive evaluation. Since VGGSound provides a single sound-event class label per clip and is typically dominated by a salient sound source with relatively clean audio-visual correspondence, an audio-visual alignment-based approach would be advantageous. Flickr-SoundNet, on the other hand, is more unconstrained and often involves ambiguous audio-visual correspondence. Thus, *EchoLoc* performs audio-conditioned object-level soft grounding, which is particularly beneficial under adaptive evaluation. This suggests that *EchoLoc* is better suited to ambiguous and multi-instance scenes, where localization requires disambiguating potential visual sound sources.

#### 3.3. Ablation study

We conduct an ablation study comparing two settings: using only the heatmap prediction module and using both the heatmap and the box prediction modules (Table 2). In this experiment, we isolate the effect of the box prediction module, while keeping other settings unchanged. Using only the heatmap prediction module, *EchoLoc* achieves the cIoU of 31.73 and the AUC of 32.06. Adding an additional box module significantly improves both metrics, increasing cIoU to 36.36(+4.63) and AUC to 38.28(+6.22).

This improvement comes from using heatmap-derived positional cues: the heatmap-only model localizes rough sound areas, and the box prediction module learns clearer object-level boundaries. This result shows that the heatmap-only model can localize rough sound-producing regions, while the box prediction module further helps the model learn object-wise boundary information, enhancing precision and consistency. In other words, the proposed box prediction module acts as a key factor in elevating heatmap-based localization to object-aware local-

Table 1: Audio-Visual Grounding Results on VGG-SS and Flickr-SoundNet Testsets

| Method                | VGG-SS       |              |              |              | Flickr-SoundNet |              |              |              |
|-----------------------|--------------|--------------|--------------|--------------|-----------------|--------------|--------------|--------------|
|                       | cIoU↑        | cIoU(adap)↑  | AUC↑         | AUC(adap)↑   | cIoU↑           | cIoU(adap)↑  | AUC↑         | AUC(adap)↑   |
| Attention [5]         | 18.50        | —            | 30.20        | —            | 66.00           | —            | 55.80        | —            |
| CoarseToFine [16]     | 29.10        | —            | 34.80        | —            | —               | —            | —            | —            |
| LCBM [17]             | 32.20        | —            | 36.60        | —            | —               | —            | —            | —            |
| LVS [6]               | 30.30        | 40.47        | 36.40        | 41.75        | 71.20           | 77.20        | 58.06        | 62.72        |
| HardPos [18]          | 34.60        | —            | 38.00        | —            | 75.20           | —            | <u>59.70</u> | —            |
| SSPL [19]             | 27.00        | —            | 34.80        | —            | 69.90           | —            | 58.00        | —            |
| EZ-VSL [7]            | 35.96        | 43.52        | 38.20        | 42.41        | 72.80           | 78.00        | 58.82        | 63.82        |
| SLAVC [20]            | 37.79        | <u>49.41</u> | <u>39.40</u> | <u>45.79</u> | —               | —            | —            | —            |
| MarginNCE [21]        | <u>38.25</u> | <b>50.76</b> | 39.06        | <b>46.39</b> | <b>84.74</b>    | <u>86.00</u> | <b>63.08</b> | <u>68.32</u> |
| FNAC [22]             | <b>39.50</b> | 47.00        | <b>39.66</b> | 43.30        | 78.71           | 84.40        | 59.33        | 64.16        |
| <b>EchoLoc (Ours)</b> | 36.36        | 44.31        | 38.28        | 41.19        | <u>84.34</u>    | <b>88.22</b> | 54.29        | <b>79.99</b> |

Table 2: Ablation results on VGG-SS, showing the impact of adding the box prediction module in EchoLoc

| Method                               | VGG-SS       |              |
|--------------------------------------|--------------|--------------|
|                                      | cIoU↑        | AUC↑         |
| Only heatmap prediction module       | <u>31.73</u> | <u>32.06</u> |
| Heatmap module + Box prediction head | <b>36.36</b> | <b>38.28</b> |

ization.

### 3.4. Discussion: Complementary audio cues under weak textual supervision

Real-world multimodal videos are often accompanied by imperfect textual signals rather than carefully curated captions. Under such weak textual supervision, text-based grounding methods that rely on explicit noun phrases become unstable; while they can predict precise object boundaries under ideal conditions, their reliability often degrades when descriptions are noisy or under-specified. In contrast, *EchoLoc* leverages audio cues to estimate sound-generating regions and produces more fine-grained soft localization in the form of heatmaps, even when textual information is insufficient. As illustrated in Figure 2 on the VGG-SS dataset, the heatmap provides a continuous spatial distribution that successfully captures audio-visual contextual consistency, instead of enforcing a single hard boundary.

These observations suggest that text-based grounding and audio-based localization provide complementary supervisory signals. While text excels at precise boundary prediction given clear linguistic instructions, audio serves as auxiliary and orthogonal cues when reliable text is difficult to obtain. Thus, *EchoLoc* complements text-based grounding by offering robust localization cues under weak textual supervision.

## 4. Conclusion

We have presented *EchoLoc*, an audio-aware grounding framework that jointly performs heatmap-based and box-level localization. On the Flickr-SoundNet benchmark, *EchoLoc* has achieved 88.22 cIoU (adap) and 79.99 AUC (adap), demonstrating the effectiveness of integrating audio-guided heatmap cues with object-centric box prediction. Our key contribution is to train a heatmap module via spatial and global contrastive learning, and use pseudo boxes generated from the predicted

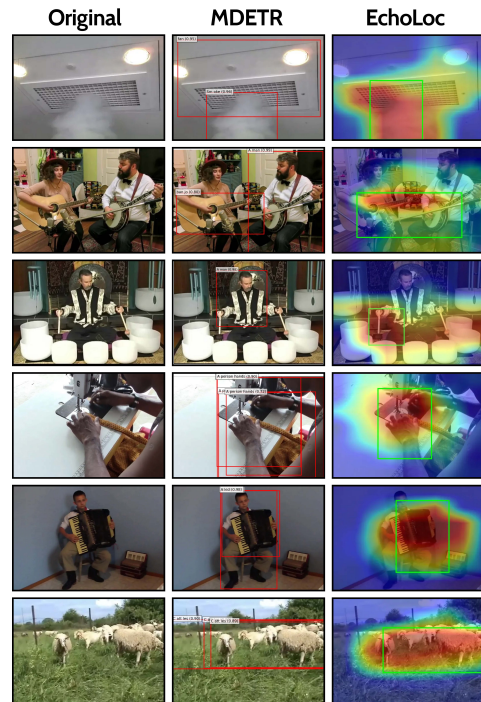


Figure 2: Qualitative comparison of sound source localization on the VGG-SS dataset under weak textual supervision. MDETR produces hard, object-centric predictions biased toward salient visual objects, whereas EchoLoc generates soft, audio-guided heatmaps.

heatmap as supervision for training the box prediction module. In this way, *EchoLoc* yields more precise object-centric boxes while also providing heatmap-based soft localization.

Despite these advantages, box-level predictions depend on the quality of the underlying heatmap estimation, since both pseudo supervision and box selection are guided by heatmap-based audio-visual similarity. As future work, we may improve the robustness of heatmap learning by incorporating additional training signals, such as explicit positive and negative audio-visual pairs explored in prior studies, and investigate strategies that can reduce the dependency of box prediction on heatmap quality.

## 5. Generative AI Use Disclosure

Generative AI tools were used solely for language editing and proofreading of this manuscript. All technical content, methodology, experiments, and conclusions were developed and verified by the authors.

## 6. References

- [1] A. Guzhov, F. Raue, J. Hees, and A. Dengel, "Audioclip: Extending clip to image, text and audio," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 976–980.
- [2] R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, "Imagebind: One embedding space to bind them all," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 15 180–15 190.
- [3] A. Kamath, M. Singh, Y. LeCun, G. Synnaeve, I. Misra, and N. Carion, "Mdetr-modulated detection for end-to-end multi-modal understanding," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 1780–1790.
- [4] L. H. Li, P. Zhang, H. Zhang, J. Yang, C. Li, Y. Zhong, L. Wang, L. Yuan, L. Zhang, J.-N. Hwang *et al.*, "Grounded language-image pre-training," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 965–10 975.
- [5] A. Senocak, T.-H. Oh, J. Kim, M.-H. Yang, and I. S. Kweon, "Learning to localize sound source in visual scenes," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 4358–4366.
- [6] H. Chen, W. Xie, T. Afouras, A. Nagrani, A. Vedaldi, and A. Zisserman, "Localizing visual sounds the hard way," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 16 867–16 876.
- [7] S. Mo and P. Morgado, "Localizing visual sounds the easy way," in *European Conference on Computer Vision*. Springer, 2022, pp. 218–234.
- [8] A. Senocak, H. Ryu, J. Kim, T.-H. Oh, H. Pfister, and J. S. Chung, "Sound source localization is all about cross-modal alignment," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 7777–7787.
- [9] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, "Panns: Large-scale pretrained audio neural networks for audio pattern recognition," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 2880–2894, 2020.
- [10] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio Spectrogram Transformer," in *Interspeech 2021*, 2021, pp. 571–575.
- [11] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *European conference on computer vision*. Springer, 2020, pp. 213–229.
- [12] H. Chen, W. Xie, A. Vedaldi, and A. Zisserman, "Vggsound: A large-scale audio-visual dataset," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 721–725.
- [13] Y. Aytar, C. Vondrick, and A. Torralba, "Soundnet: Learning sound representations from unlabeled video," *Advances in neural information processing systems*, vol. 29, 2016.
- [14] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations*, 2019. [Online]. Available: <https://openreview.net/forum?id=Bkg6RiCqY7>
- [15] A. Senocak, H. Ryu, J. Kim, T.-H. Oh, H. Pfister, and J. S. Chung, "Toward interactive sound source localization: Better align sight and sound!" *IEEE transactions on pattern analysis and machine intelligence*, 2025.
- [16] R. Qian, D. Hu, H. Dinkel, M. Wu, N. Xu, and W. Lin, "Multiple sound sources localization from coarse to fine," in *European Conference on Computer Vision*. Springer, 2020, pp. 292–308.
- [17] A. Senocak, H. Ryu, J. Kim, and I. S. Kweon, "Less can be more: Sound source localization with a classification model," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 3308–3317.
- [18] —, "Learning sound localization better from semantically similar samples," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 4863–4867.
- [19] Z. Song, Y. Wang, J. Fan, T. Tan, and Z. Zhang, "Self-supervised predictive learning: A negative-free method for sound source localization in visual scenes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 3222–3231.
- [20] S. Mo and P. Morgado, "A closer look at weakly-supervised audio-visual source localization," *Advances in Neural Information Processing Systems*, vol. 35, pp. 37 524–37 536, 2022.
- [21] S. Park, A. Senocak, and J. S. Chung, "Marginnce: Robust sound localization with a negative margin," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [22] W. Sun, J. Zhang, J. Wang, Z. Liu, Y. Zhong, T. Feng, Y. Guo, Y. Zhang, and N. Barnes, "Learning audio-visual source localization via false negative aware contrastive learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 6420–6429.